

Restaurant Rating Prediction

1. Introduction

Restaurant ratings play an important role in helping customers choose restaurants and helping businesses understand customer satisfaction. The objective of this project is to build a machine learning model that predicts the aggregate rating of a restaurant based on various restaurant attributes.

2. Objective

To develop and evaluate machine learning models capable of predicting restaurant ratings using features such as location, cuisine type, cost, online delivery options, and customer votes.

3. Dataset Description

The dataset contains information about restaurants, including their location, cuisines, pricing, customer engagement, and ratings.

Dataset Information

- Total Records: 9551
- Total Features: 21
- Target Variable: Aggregate Rating

Important Features Used

- Country Code
 - City
 - Cuisines
 - Average Cost for Two
 - Currency
 - Has Table Booking
 - Has Online Delivery
 - Is Delivering Now
 - Switch to Order Menu
 - Price Range
 - Votes
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4. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the dataset using Pandas.
2. Checked for missing values.
3. Removed 9 records with missing values in the Cuisines column.
4. Encoded categorical variables using Label Encoding.
5. Selected relevant features for prediction.
6. Split the dataset into training (80%) and testing (20%) sets.

Training Data Shape: (7633, 11)

Testing Data Shape: (1909, 11)

5. Exploratory Data Analysis (EDA)

Missing Values

Only the Cuisines column contained missing values (9 records), which were removed.

Rating Distribution

The distribution of restaurant ratings showed that most restaurants have ratings between 2.5 and 4.5, while a significant number of restaurants have ratings close to zero.

Figure 1: Distribution of Restaurant Ratings

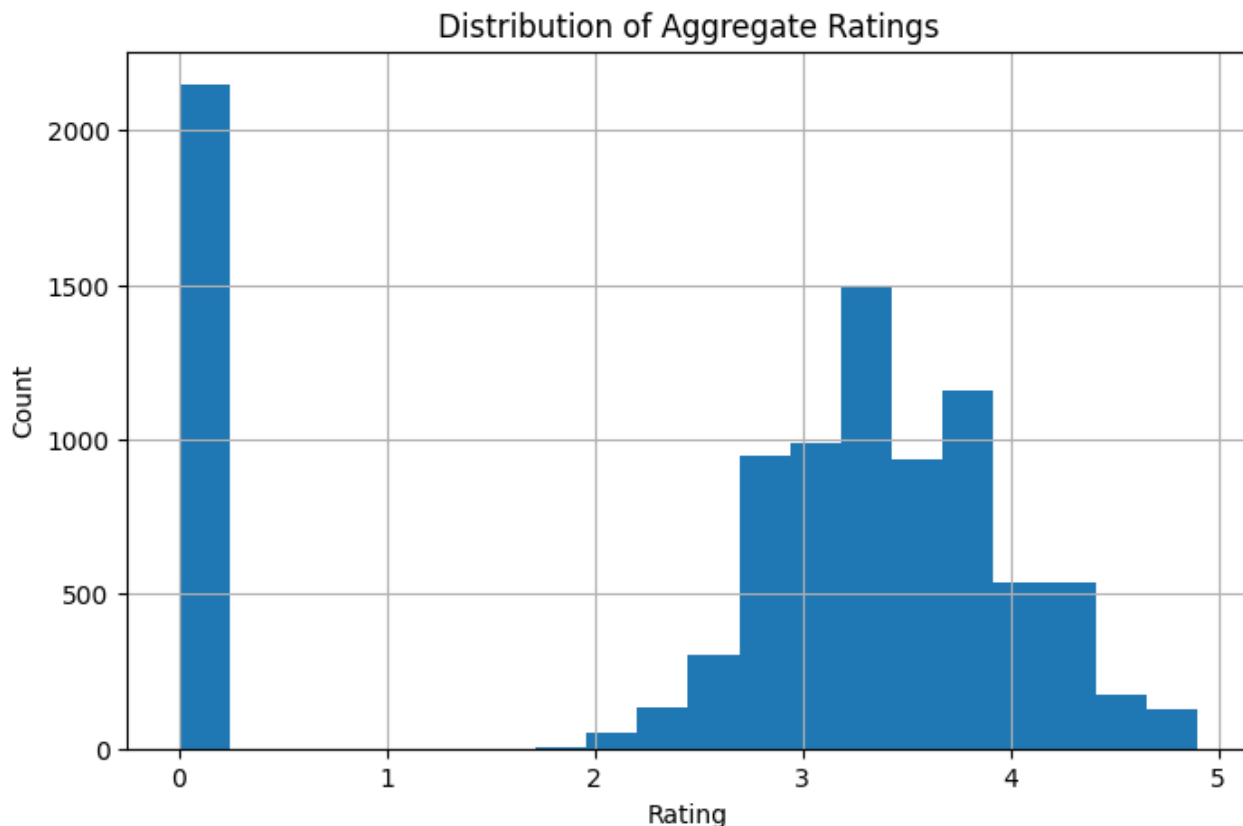


Figure 1

shows that most restaurants have ratings in the range of 2.5 to 4.5, indicating that the majority of restaurants receive moderate to high customer ratings.

Top Cities

- New Delhi
- Gurgaon
- Noida
- Faridabad
- Ghaziabad

The dataset is heavily concentrated in the Delhi NCR region.

Top Cuisines

- North Indian
- North Indian, Chinese
- Chinese
- Fast Food
- North Indian, Mughlai

Correlation Analysis

The strongest correlations with Aggregate Rating were:

- Price Range: 0.438
- Votes: 0.313

These features showed a positive relationship with restaurant ratings.

6. Model Development

Three regression algorithms were implemented and evaluated:

1. Linear Regression

A baseline regression model used to establish initial performance.

2. Decision Tree Regressor

A tree-based model capable of capturing non-linear relationships in the dataset.

3. Random Forest Regressor

An ensemble learning method combining multiple decision trees to improve prediction accuracy and reduce overfitting.

7. Model Evaluation

Performance Comparison

Model	Mean Squared Error (MSE)	R ² Score
Linear Regression	1.5242	0.3344
Decision Tree Regressor	0.1738	0.9241
Random Forest Regressor	0.0993	0.9566

Best Model

The Random Forest Regressor achieved the highest performance with:

- MSE = 0.0993
- R² Score = 0.9566

This indicates that the model can explain approximately 95.66% of the variation in restaurant ratings.

8. Feature Importance Analysis

Figure 2: Feature Importance from Random Forest Model

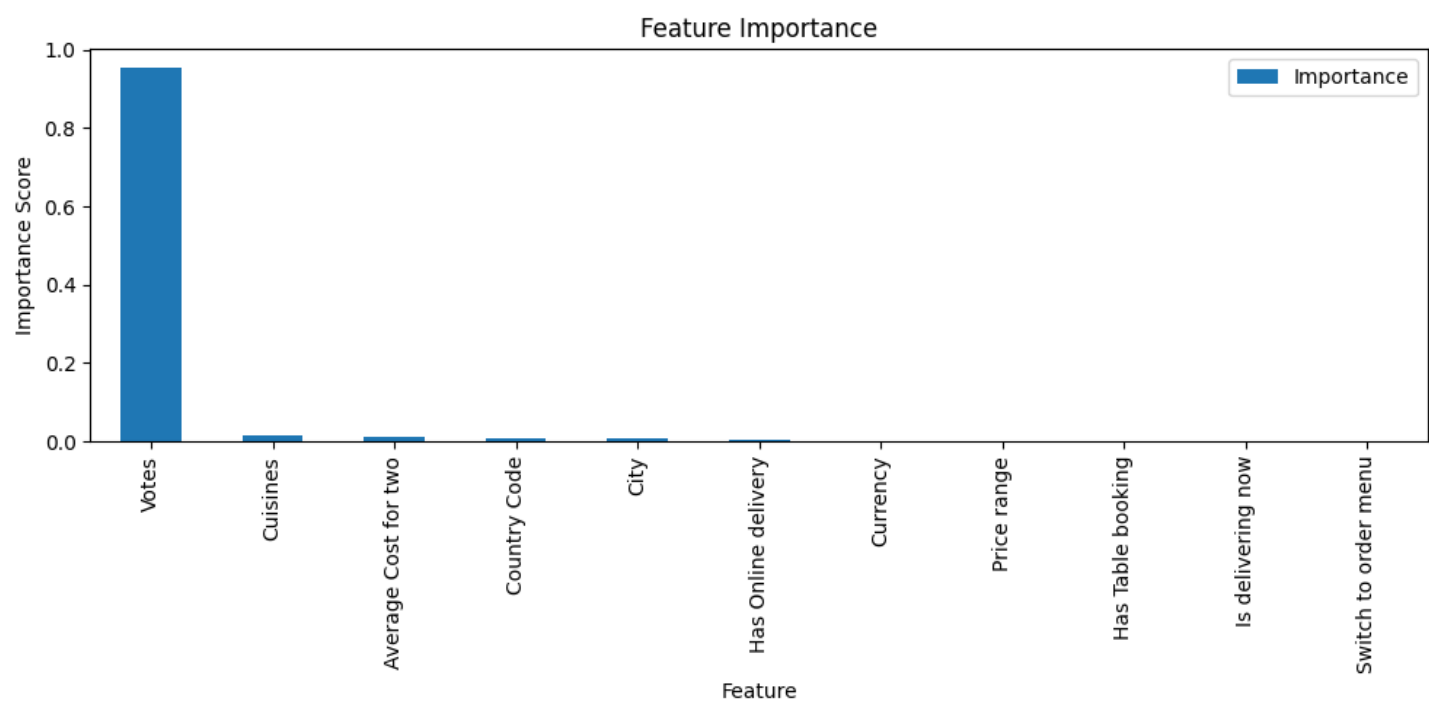


Figure 2 illustrates that Votes is the most influential feature affecting restaurant ratings, contributing significantly more than all other features.

Feature importance obtained from the Random Forest model:

Feature	Importance
Votes	0.9540
Cuisines	0.0155
Average Cost for Two	0.0094
Country Code	0.0082
City	0.0072

Observations

- Votes was the most influential feature affecting restaurant ratings.
 - Cuisine type contributed moderately to rating prediction.
 - Restaurant location and pricing had smaller effects.
 - Features such as table booking and online delivery had minimal influence.
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9. Conclusion

A machine learning system was successfully developed to predict restaurant ratings using restaurant attributes and customer engagement data.

Three regression models were evaluated, and the Random Forest Regressor achieved the best performance with an R^2 score of 0.9566 and an MSE of 0.0993.

The analysis revealed that customer votes are the most significant factor influencing restaurant ratings. The developed model can effectively predict restaurant ratings and may assist businesses in understanding the factors that contribute to customer satisfaction.

10. Tools and Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-learn
- Jupyter Notebook
- Visual Studio Code